Reflect with us - Week 6

This assignment will not be graded and is only for practice.
Suppose the matrix representation of a linear transformation $T: R^3 \rightarrow R^3$ T : R3 → R3 with respect to ordered bases

$\beta = \{(1,0,1),(0,1,0),(0,0,1)\}$ β = {(1, 0, 1), (0, 1, 0), (0, 0, 1)} for the domain and

$\gamma = \{(1,0,0),(0,1,0),(1,0,1)\}$ γ = {(1, 0, 0), (0, 1, 0), (1, 0, 1)} for the range,

is $I_{3\times 3}$ I3×3, i.e., the identity matrix of order 3.

Let A A denote the matrix representation of the linear transformation T T with respect to the standard ordered basis of R^3 R3 for both domain and range. Which of the following are true?

a. $A = I_{3\times 3}$ A = I3×3 i.e., identity matrix of order 3.
b. A A is a singular matrix.
c. $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & -1 \end{bmatrix}$ $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$
d. $det(A)$ = 1det(A) = 1
e. $det(A) = -1$ det(A) = −1
f. $\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$ $\begin{bmatrix} 0 & 0 & -1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$

Solution:

Observe that, in question, you are asked to find the matrix representation of T T with respect to the standard ordered bases of R^3 R3 for both domain and range. So you have to find the image of each element of the standard ordered basis of R^2 R2 via the linear transformation T T.

Note:Note: $I_{3\times 3}$ I3×3 is matrix representation of the T T with respect to the ordered bases β β and γ γ.

Step 0:Step 0:

What is $T(1,0,1)$ T(1, 0, 1)? (Enter your answer in the form of a row vector in R^3 R3)

Note: The answer will be of the form (a, b, c) where a,b and c are in R R

No, the answer is incorrect.
Score: 0
Feedback:
From the given data, we can calculate
$T(1,0,1) = 1(1,0,0) + 0(0,1,0) + 0(1,0,1)$ T(1, 0, 1) = 1(1, 0, 0) + 0(0, 1, 0) + 0(1, 0, 1)

Accepted Answers:

(Type: String) (1, 0, 0)
(Type: String) (1,0,0)
1 point

What is $T(0, 1, 0)$T(0, 1, 0)?

No, the answer is incorrect.
Score: 0
Feedback:
From the given data, we can calculate
$T(0, 1, 0) = 0(1, 0, 0) + 1(0, 1, 0) + 0(1, 0, 1)$T(0, 1, 0) = 0(1, 0, 0) + 1(0, 1, 0) + 0(1, 0, 1)
Accepted Answers:
(Type: String) (0, 1, 0)
(Type: String) (0,1,0)
1 point

What is $T(0, 0, 1)$T(0, 0, 1)?

No, the answer is incorrect.
Score: 0
Feedback:
From the given data, we can calculate
$T(0, 0, 1) = 0(1, 0, 0) + 0(0, 1, 0) + 1(1, 0, 1)$T(0, 0, 1) = 0(1, 0, 0) + 0(0, 1, 0) + 1(1, 0, 1)
Accepted Answers:
(Type: String) (1, 0, 1)
(Type: String) (1,0,1)
1 point

Step 1: Step 1:
Now we have to find the image of $(1, 0, 0)$(1, 0, 0) via the linear transformation TT. We know the image of each element of the basis $\{(1, 0, 1), (0, 1, 0), (0, 0, 1)\}$\{(1, 0, 1), (0, 1, 0), (0, 0, 1)\}. So at first we have to write $(1, 0, 0)$(1, 0, 0) in terms of the elements of the basis $\{(1, 0, 1), (0, 1, 0), (0, 0, 1)\}$\{(1, 0, 1), (0, 1, 0), (0, 0, 1)\}.

$$(1, 0, 0) = \alpha_1 (1, 0, 1) + \beta_1 (0, 1, 0) + \gamma_1 (0, 0, 1)(1, 0, 0) = \alpha 1(1, 0, 1) + \beta 1(0, 1, 0) + \gamma 1$$

(0, 0, 1)
What is the value of α_1 α1?

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 1
1 point

What is the value of β_1 β1?

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 0
1 point

What is the value of γ_1 γ1?

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) -1
1 point

1 point
Now, can you find $T(1,0,0)$T(1,0,0)?

- [ ] Yes
- [ ] No

No, the answer is incorrect.
Score: 0
Feedback:
From the given data, we can calculate
$T(1,0,0) = T(1(1,0,1) + 0(0,1,0) - 1(0,0,1))T(1,0,0) = 1T(1,0,1) + 0T(0,1,0) - 1T(0,0,1)$
T(1,0,0) = T(1(1,0,1) + 0(0,1,0) − 1(0,0,1))T(1,0,0) = 1T(1,0,1) + 0T(0,1,0) − 1T(0,0,1)

Here we know the images of $(1,0,1),(0,1,0)$(1,0,1),(0,1,0) and $(0,0,1)$.(0,0,1).
$T(1,0,0) = 1(1,0,0) + 0(0,1,0) - 1(1,0,1) = (0,0,-1)$T(1,0,0) = 1(1,0,0) + 0(0,1,0) − 1(1,0,1) = $(0,0,-1)$

Accepted Answers:
Yes
No

Step 2:Step 2:
Now we have to find the image of $(0,1,0)$(0,1,0) via the linear transformation TT. We know the image of each element of the basis $\{(1,0,1),(0,1,0),(0,0,1)\}$\{(1,0,1),(0,1,0),(0,0,1)\}. So at first we have to write $(0,1,0)$(0,1,0) in terms of the elements of the basis $\{(1,0,1),(0,1,0),(0,0,1)\}$\{(1,0,1),(0,1,0),(0,0,1)\}.

$(0,1,0) = \alpha_2(1,0,1) + \beta_2(0,1,0) + \gamma_2(0,0,1)$(0,1,0) = α2(1,0,1) + β2(0,1,0) + γ2(0,0,1)

What is the value of α_2 α2?

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 0
1 point

What is the value of β_2 β2?

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 1

1 point

What is the value of γ_2 γ2?

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 0
1 point

1 point
Now, can you find $T(0,1,0)$T(0, 1, 0)?

- [ ] Yes
- [ ] No

No, the answer is incorrect.
Score: 0
Feedback:
$T(0,1,0) = T(0(1,0,1) + 1(0,1,0) + 0(0,0,1))$T(0, 1, 0) = T(0(1, 0, 1) + 1(0, 1, 0) + 0(0, 0, 1))
$T(0,1,0) = 0T(1,0,1) + 1T(0,1,0) + 0T(0,0,1)$T(0, 1, 0) = 0T(1, 0, 1) + 1T(0, 1, 0) + 0T(0, 0, 1)
$T(0,1,0) = (0,1,0)$T(0, 1, 0) = (0, 1, 0)
Accepted Answers:
Yes
No

Step 3:Step 3:
Now we have to find the image of $(0,0,1)$(0, 0, 1) via the linear transformation TT. We know the image of each element of the basis $\{(1,0,1),(0,1,0),(0,0,1)\}$\{(1, 0, 1), (0, 1, 0), (0, 0, 1)\}. So at first we have to write $(0,0,1)$(0, 0, 1) in terms of the elements of the basis $\{(1,0,1),(0,1,0),(0,0,1)\}$\{(1, 0, 1), (0, 1, 0), (0, 0, 1)\}.
$(0,0,1) = \alpha_3(1,0,1) + \beta_3(0,1,0) + \gamma_3(0,0,1)$(0, 0, 1) = α3(1, 0, 1) + β3(0, 1, 0) + γ3(0, 0, 1)
What is the value of α_3 α3?

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 0
1 point

What is the value of β_3 β3?

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 0
1 point

What is the value of γ_3 γ3?

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 1
1 point

1 point
Now, can you find $T(0,0,1)$T(0, 0, 1)?
- [ ] Yes
- [ ] No

No, the answer is incorrect.
Score: 0
Feedback:
$T(0,0,1) = T(0(1,0,1) + 0(0,1,0) + 1(0,0,1))$T(0, 0, 1) = T(0(1, 0, 1) + 0(0, 1, 0) + 1(0, 0, 1))
$T(0,0,1) = 0T(1,0,1) + 0T(0,1,0) + 1T(0,0,1)$T(0, 0, 1) = 0T(1, 0, 1) + 0T(0, 1, 0) + 1T(0, 0, 1)
$T(0,0,1) = (1,0,1)$T(0, 0, 1) = (1, 0, 1)

Accepted Answers:
Yes
No

Step 4:Step 4: Now we have to find out the matrix AA. For that we have to write down the images with respect to the standard ordered basis.

$T(1,0,0) = 0\ (1,0,0) + 0\ (0,1,0) - 1\ (0,0,1)$T(1, 0, 0) = 0 (1, 0, 0) + 0 (0, 1, 0) − 1 (0, 0, 1)
$T(0,1,0) = 0\ (1,0,0) + 1\ (0,1,0) + 0\ (0,0,1)$T(0, 1, 0) = 0 (1, 0, 0) + 1 (0, 1, 0) + 0 (0, 0, 1)
$T(0,0,1) = 1\ (1,0,0) + 0\ (0,1,0) + 1\ (0,0,1)$T(0, 0, 1) = 1 (1, 0, 0) + 0 (0, 1, 0) + 1 (0, 0, 1)
Write the coefficients column wise to get the matrix AA.

$A = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$A = $\begin{bmatrix} & & \\ & & \\ 0 & 0 & -1 & 0 & 1 & 0 & 1 & 0 & 1 \end{bmatrix}$

Step 5:Step 5:
In this step, you have to find the determinant of the matrix AA.
1 point
Can you find the $det(A)$det(A)?
- [ ] Yes
- [ ] No

No, the answer is incorrect.
Score: 0
Accepted Answers:
Yes
No

- Expanding with respect to the first row:
  $det(A) = 0(1-0) - 0(0-0) + 1(0+1)$det(A) = 0(1 − 0) − 0(0 − 0) + 1(0 + 1)
- Expanding with respect to the first column:
  $det(A) = 0(1-0) - 0(0-0) - 1(0-1)$det(A) = 0(1 − 0) − 0(0 − 0) − 1(0 − 1)

- You can expand with respect to any other rows or column.
  Enter the value of $determinant\ of\ the\ matrix\ A$determinant of the matrix A

  No, the answer is incorrect.
  Score: 0
  Accepted Answers:
  (Type: Numeric) 1
  1 point
  Check Answers
  Your score is: 0/17